

AMENDMENT NO. 1 OCTOBER 2025
TO
IS 17550 (PART 3) : 2025 HOUSEHOLD REFRIGERATING APPLIANCES —
METHODS OF TEST
PART 3 ENERGY CONSUMPTION AND VOLUME
(First Revision)

(Page 5, clause 6.8.2) — Substitute the following for the existing:

‘6.8.2 Daily Energy Consumption

All values of energy consumption and power shall be converted to daily energy consumption values in accordance with the following equations for each temperature control setting and ambient temperature.

For refrigerating appliances without a defrost control cycle, the daily energy consumption for each ambient temperature and each temperature control setting is given by:

$$E_{daily} = P \times 24 \quad (1)$$

where

E_{daily} = energy in Wh over a period of 24 h;

24 = h/d; and

P = steady state power in watt for the selected temperature control setting as per Annex B.

The measured steady state temperature for each compartment shall be recorded with this value (for the test report and/or for interpolation).

For refrigerating appliances with one defrost system (with its own defrost control cycle), the daily energy consumption for each ambient temperature and each temperature control setting is based on the steady state power consumption as determined in accordance with Annex B, the incremental defrost and recovery energy determined in accordance with Annex C and the defrost interval determined in accordance with Annex D as follows:

$$E_{daily} = P \times 24 + \frac{\Delta E_{df} \times 24}{\Delta t_{df}} \quad (2)$$

where

E_{daily} = energy in Wh, over a period of 24 h;

24 = h/d;

P = steady state power in watt for the selected temperature control setting as per Annex B;

ΔE_{df} = representative incremental energy for defrost and recovery, in Wh in accordance with Annex C (see C-5); and

Δt_{df} = estimated defrost interval, in hours in accordance with Annex D.

Where there are additional defrost systems (each with its own defrost control cycle), the value of term based on ΔE_{df} and Δt_{df} is also added in equation 2 for each additional defrost system.

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The average temperature for each compartment for this temperature control setting and energy consumption is given by:

$$T_{average} = T_{ss} + \frac{\Delta Th_{df}}{\Delta t_{df}} \quad (3)$$

where

- $T_{average}$ = average temperature for the compartment over a complete defrost control cycle;
- T_{ss} = average steady state temperature in the compartment for the temperature control setting in °C in accordance with Annex B;
- ΔTh_{df} = representative accumulated temperature difference over time for defrost and recovery (relative to the steady state temperature) in degree Kelvin-hour (Kh) for the relevant compartment in accordance with Annex C (*see C-5*); and
- Δt_{df} = estimated defrost interval in hours in accordance with Annex D.

The value of ΔTh_{df} may be positive (if the temperature is warmer during defrost and recovery) or negative (if it is cooler, due to a pre-cool and low heat leakage during defrost).

Where there are additional defrost systems (each with its own defrost control cycle), the value of term based on ΔTh_{df} and Δt_{df} is also added in equation 3 for each additional defrost system.'